Questionnaire on Meta-Structure Explainability

Introduction

Welcome to participate in this questionnaire! The goal of this questionnaire is to assess the explainability of meta-structures on a heterogeneous information network to facilitate downstream tasks including recommendation and node classification.

We will first introduce the basic notions, and ask you to complete two multiple-choice questions. The whole process is expected to be completed in no more than 5 minutes.

Please read the instructions carefully before responding.

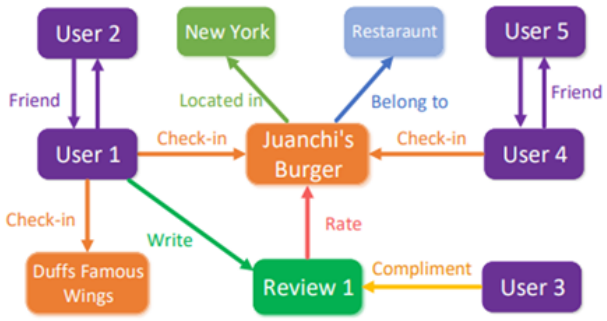
Section 0: Basic Notion

Heterogeneous Information Networks (HIN): Illustrated in Figure (a), an HIN is a network whose nodes and/or edges belong to multiple types. Figure 1 shows an example extracted from Yelp, where node types include User (U), Business (B), Review (O), City (I), and Business Categories (A), and edges connecting these nodes represent various meanings including 'Located In', 'Check-in', 'Friend', etc.

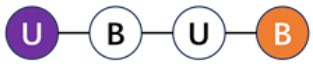
Meta-path: A meta-path is a predefined sequence of node types and edge types in an HIN, connecting a source node and a target node. These paths define the context and semantics of relationships between nodes, and serve as a filtering template for message passing, thus providing a way to incorporate higher-level structural information into machine-learning models. In Figure (b), a meta-path from a User node to a Business node reflects the collaborative filtering mechanism in recommendation systems, suggesting users' interest in businesses visited by others with similar preferences.

Meta-structure: Generalizing a meta-path, a meta-structure allows graph-like structures to exist between the source node and the target node to carry more specific meanings. In Figure (c), we introduce a meta-structure by adding another meta-path (U-U-B) to the example in Figure (b). This expanded structure considers not only business similarities, but also similar interests shared by friends, enriching the recommendation process.

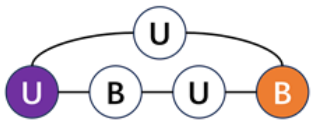
Alternative Meta-structure representation: Figure (d) presents an alternative description of the above-mentioned meta-structure using intermediate states in the message-passing process. Ordered nodes denote intermediate representations, with H^(0) being the input and H^(K) (here K=3) the output. From H^(i) to H^(i+1), the model will choose one type of edge for information propagation.



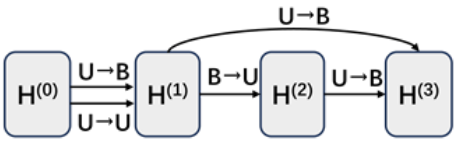
(a) Example of a Heterogeneous Information Network



(b) Example of a meta-path



(c) Example of a meta-structure



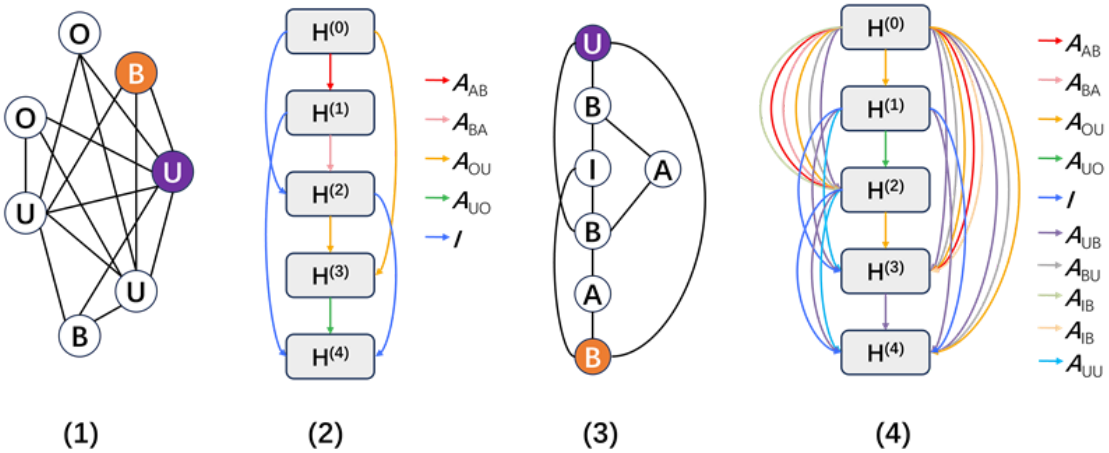
(d) Example of an alternative expression

Section 1: Structure Selection

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Below are some meta-structures generated by different models. Kindly select one meta-structure from the following options that you find most informative or comprehendible.

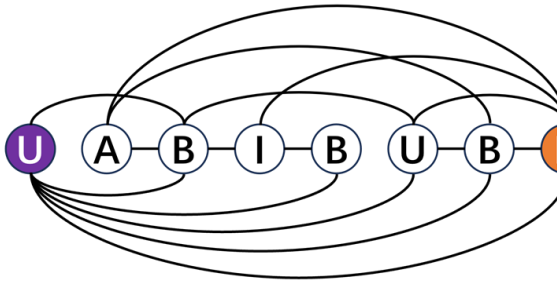
U – User, **B** – Business, **O** – Review, **I** – City, and **A** – Business Category
UB – User visits Business, **BA** – Business belongs to Business Category,
UO – User leaves a Review, **BI** – Business resides in City



(1) (2) (3) (4)

- ◯ Meta-structure (1)
- ◯ Meta-structure (2)
- ◯ Meta-structure (3)
- ◯ Meta-structure (4)

Section 2: Explanation Evaluation (1)



The figure shows a meta-structure on the Yelp network.

Node types: U – User, B – Business, O – Review, I – City, and A – Business Category
Edge types: UB – User visits Business, BA – Business belongs to Business Category, UO – User leaves a Review, BI – Business resides in City

We use the LLM to generate two explanations of why it performs well. Please read the two explanations below and decide which one is more helpful to you by improving your (1) understanding of how this meta-structure is constructed, and (2) insights on how to design a better one.

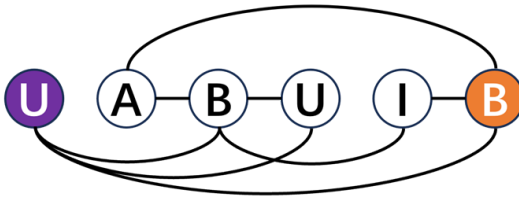
[Explanation 1]
We can identify the sub-structures that contribute positively to the performance of the given meta-structure and those that may detract from it. Here are the findings:
1. **Includes: U-B-I-B**: Captures the geographic influence on user preferences, indicating that users tend to visit businesses within a certain location or city, which is a strong predictor of user behavior.
2. **Includes: U-U-B**: Represents the impact of social relationships on user behavior, showing that users are influenced by the places their friends visit. This social proof is a key factor in recommendations.
3. **Includes: U-B-A-B**: Highlights the importance of business categories in user preferences. Users who visit one business are likely to visit other businesses within the same category, which can be used to recommend similar businesses.
4. **Not includes: U-B-A-B-U-U**: Appearing in some one-step neighbors with lower performance, this sub-structure adds complexity by connecting user preferences with a friend's visit to a business in the same category. This complexity does not seem to improve the model's performance and could potentially overfit the recommendation algorithm.
5. **Not includes: U-B-A-B-A**: Introduces redundancy in business categories, suggesting a business belongs to multiple categories or a chain of category relationships. This redundancy does not improve performance and may introduce noise into the recommendation model.
In summary, the given meta-structure performs well by incorporating effective sub-structures that capture location-based preferences, social influence, and category interests. Importantly, it avoids unnecessary complexity or redundancy that could compromise the model's predictive power.
By analyzing the performances and structures jointly, we conclude that the given meta-structure performs well because it includes effective sub-structures that capture location-based preferences, social influence, and category interests without introducing unnecessary complexity or redundancy that could detract from the model's predictive power.

[Explanation 2]
The meta-structure captures the influence of direct user preferences by including paths where a user (U) visits various local businesses (B), suggesting that the businesses a user directly interacts with are strong predictors of their future interests.- It incorporates the social aspect of user behavior by including user friendships (U-U) and the consequent overlap in businesses visited by both the user and their friends, leveraging the idea that friends often share similar tastes and can influence each other's choices.- The meta-structure utilizes business categories (A) to connect businesses visited by the same user or their friends, highlighting the significance of categorical preference in recommendations, as users are likely to visit different businesses within the same category.- It takes into account the geographical context by involving business cities (I), which can be a deciding factor for users when choosing local businesses, thus providing location-based personalization in the recommendations.- By combining these elements, the meta-structure effectively integrates user behavior, social connections, categorical preferences, and geographical context to provide a holistic view that contributes to its high performance in a recommendation model.

◯ Explanation 1

◯ Explanation 2

Section 2: Explanation Evaluation (2)



The figure shows a meta-structure on the Yelp network.

Node types: U – User, B – Business, O – Review, I – City, and A – Business Category
Edge types: UB – User visits Business, BA – Business belongs to Business Category, UO – User leaves a Review, BI – Business resides in City

We use the LLM to generate two explanations of why it performs well. Please read the two explanations below and decide which one is more helpful to you, by improving your (1) understanding of how this meta-structure is constructed, and (2) insights on how to design a better one.

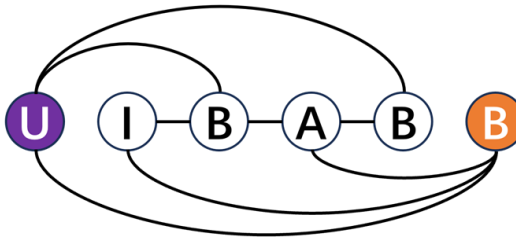
[Explanation 1]
let's identify the sub-structures that contribute positively (good sub-structures) to the performance of the given meta-structure, as well as those that are absent and potentially detrimental (bad sub-structures).
1. Includes: U-B: Direct user-business interactions, fundamental to indicating the user's personal preference in recommendation systems.
2. Includes: U-B-A-B: Adds context to the user-business visits through a shared category. It indicates that users who visit businesses within the same category are more likely to have similar tastes, which is key for category-based recommendations.
3. Includes: U-U: Captures social relationships between users, which is a strong indicator of shared preferences. It's important because friends often have similar tastes, and their preferences can significantly influence a user's decisions.
4. Not includes: U-U-B: Fails to significantly improve performance, suggesting that direct friendship connection to the target business is less influential.
5. Not includes: U-O: Absent in the given meta-structure, indicating compliments may not be a critical indicator of business preference.
6. Not includes: U-B-U-B: Absence and relative performance decrease suggest this additional layer of social influence may not be effective.
7. Not includes: U-B-...-I-B: Present but unclear if its absence would be detrimental. A neighboring meta-structure shows a slight performance decrease, suggesting added geographical complexity may not always be beneficial.
In summary, direct user-business interactions, category-based connections, and social relationships are strong indicators for successful recommendations.

[Explanation 2]
The given meta-structure performs well because it captures several important aspects of user behavior and business characteristics that are relevant for recommendations. Firstly, it considers direct visits by the user to various businesses, indicating personal preferences. Secondly, it incorporates social influence by including businesses visited by the user's friends, which suggests that recommendations can be based on social circles and shared interests. Thirdly, it accounts for common business categories, implying that users are likely to be interested in other businesses within the same category as those they have visited before. Fourthly, it includes other users who have visited the same businesses, allowing for collaborative filtering based on the behavior of similar users. Lastly, it takes into account the geographical context by including the city where the businesses are located, which is a practical factor as users often prefer local businesses. This rich combination of user preferences, social influence, categorical similarity, collaborative filtering, and geographical context makes the meta-structure particularly effective for personalized business recommendations.

◯ Explanation 1

◯ Explanation 2

Section 2: Explanation Evaluation (3)



The figure shows a meta-structure on the Yelp network.

Node types: U – User, B – Business, O – Review, I – City, and A – Business Category
Edge types: UB – User visits Business, BA – Business belongs to Business Category, UO – User leaves a Review, BI – Business resides in City

We use the LLM to generate two explanations of why it performs well. Please read the two explanations below and decide which one is more helpful to you, by improving your (1) understanding of how this meta-structure is constructed, and (2) insights on how to design a better one.

4

[Explanation 1]
The given meta-structure performs well because it captures multiple dimensions of user preferences and business characteristics. It considers not only direct visits by the user to various businesses but also contextual information such as the geographical location of businesses and their categories. By including businesses visited by the user that are in the same location or category as another visited business, it leverages the assumption that users may prefer businesses within certain locales or of specific types. Additionally, the structure accounts for the user's repeated interest in a business category, as indicated by visits to multiple businesses within the same category. This multi-faceted approach provides a rich and nuanced view of user behavior and preferences, which is likely why it yields a strong performance in the recommendation model.

[Explanation 2]
We can identify the sub-structures that positively or negatively contribute to the given meta-structure's performance:
1. Includes: U-B-A-B: Captures user preference for a specific business category (A), a strong indicator of the user's interests and is likely to improve recommendation quality.
2. Includes: U-B-I-B: Incorporates user preference for businesses within a particular city or geographical area, highlighting the importance of location in user preferences and strengthening the recommendation model.
3. Not includes: U-U-B: Absence in some of the better-performing neighbors, indicating that direct friendship connections may be less influential than factors like business categories or geographical location, in determining user preferences for recommendations.
4. Includes: U-B-B-I-A: This complex sub-structure includes both geographical and categorical preferences, which seems to perform very well, suggesting that a combination of these factors is highly informative for recommendations.
5. Not includes: U-B-U-U: This sub-structure, which emphasizes social connections between users based on shared visits, is absent from the given meta-structure. The absence of this sub-structure might indicate that the social aspect is not as critical when other strong indicators, such as category and location, are present.
In summary, the given meta-structure performs well because it effectively captures the user's categorical and geographical preferences, which are stronger predictors for recommendations compared to the social connections between users. The absence of certain sub-structures like direct friendship connections suggests their relative insignificance in the context of this particular Yelp HIN for recommendations.

◯ Explanation 1

◯ Explanation 2

Section 3: Additional Feedback

5

If you have any additional comments or suggestions regarding the clarity or effectiveness of the explanations or the questionnaire itself, please provide them below. Thank you again for your valuable input!

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